

7.4 Exercises: Understanding and inquiring

To answer questions online and to receive **immediate feedback** and **sample responses** for every question, go to your learnON title at www.jacplus.com.au. *Note:* Question numbers may vary slightly.

Remember

1. Suggest why Earth's atmosphere has been described as a giant invisible blanket.
2. What is:
 - (a) the greenhouse effect
 - (b) the enhanced greenhouse effect
 - (c) global warming?
3. Suggest four consequences of global warming.
4. Give examples of three types of greenhouse gases and at least two sources for each.
5. Identify the links between photosynthesis, cellular respiration, decomposition, fossil fuels and global warming.
6. Explain why ozone in the Earth's stratosphere is important to humans and all other life on Earth.
7. Explain how scientists are able to determine the air temperature and the amount of carbon dioxide in the atmosphere hundreds of thousands of years ago when such measurements were never recorded.
8. Explain how the thawing of permafrost could increase the rate of global warming.

Think and discuss

9. Outline the actions that individuals can take to slow the rate of global warming.
10. (a) In your own words, describe what is meant by the term *enhanced greenhouse effect*.
(b) Suggest a model or simulation that could communicate this concept to others.
11. Suggest how whales that live on plankton could be affected by global warming.
12. (a) Which of the following actions would you be prepared to take so that you can contribute to the fight against global warming?
 - Walk, cycle or use public transport rather than relying on someone to drive you to school, work or leisure activities.
 - Change your diet so that you eat less meat and more fruit and vegetables.
 - Recycle paper, aluminium and steel cans, glass and plastics.
 - Stop using electric clothes dryers and use outdoor clothes lines in dry weather and indoor folding clothes-airers in wet weather to dry clothes.(b) Select one of the actions in part (a) that the government could enforce by passing new laws and explain how it could be done.
13. Explain why the average temperature of the Earth's atmosphere was constantly changing for millions of years before humans existed.
14. Outline the likely effect on land-based living things caused by:
 - (a) rising sea levels
 - (b) an increase in average temperatures
 - (c) significantly increased rainfall
 - (d) significantly decreased rainfall.
15. Explain why it is necessary for the Australian government to create legislation to address the problem of global warming.